

EV → PP

Query

Explanation Segment (Summary): Discusses about attention analysis in detail.

Candidate 1

ABSTRACT

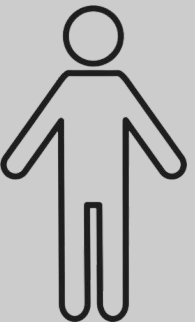
Transformer architectures have proven to learn useful representations for protein classification and generation tasks. However, these representations present challenges in interpretability. In this work, we demonstrate a set of methods for analyzing protein Transformer models through the lens of attention. We show that attention: (1) captures the folding structure of proteins, connecting amino acids that are far apart in the underlying sequence, but spatially close in the three-dimensional structure, (2) targets binding sites, a key functional component of proteins, and (3) focuses on progressively more complex biophysical properties with increasing layer depth. We find this behavior to be consistent across three Transformer architectures (BERT, ALBERT, XLNet) and two distinct protein datasets. We also present a three-dimensional visualization of the interaction between attention and protein structure. Code for visualization and analysis is available at <https://github.com/salesforce/provis>.

OR

Candidate 2

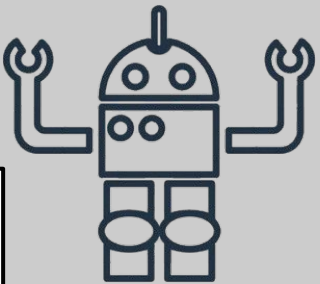
Attention analysis. We analyze how attention aligns with various protein properties. For properties of token pairs, e.g. contact maps, we define an indicator function $f(i, j)$ that returns 1 if the property is present in token pair (i, j) (e.g., if amino acids i and j are in contact), and 0 otherwise. We then compute the proportion of high-attention token pairs ($\alpha_{i,j} > \theta$) where the property is present, aggregated over a dataset $\mathbf{X}$:

Human



Candidate 2 is fine-grain match

VLM



Candidate 1: 0.95
Candidate 2: 0.9
Choice: Candidate 1

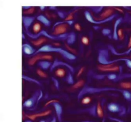
S → PP

Query

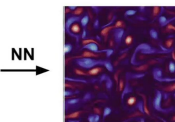
Background - Neural PDE solver

Using neural networks to predict the solution function of partial differential equations.

$u = \mathcal{G}(a)$, with a being input function and u is the target



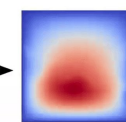
Input: $\omega(\cdot, t)$



Output: $\omega(\cdot, t + \Delta t)$



Input: a



Output: u

Candidate 1

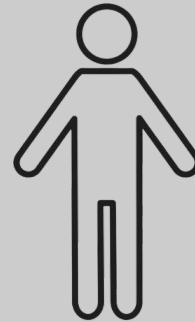
Neural PDE solver Based on the emphases of model design, neural PDE solvers can generally be divided into the following groups. The first group of work focuses on using neural networks with mesh-specific architecture design (such as convolutional layers for uniform mesh, or a graph layer for irregular mesh) to learn the spatial and/or temporal correlation of the PDE data [10, 38, 48, 64, 65, 75, 88, 91, 92, 100, 104, 109, 114, 116]. With input-target data collected, the training process can be conducted without the knowledge of underlying PDEs. This can be appealing when the physics of the system is unknown or partially known, such as large-scale climate modeling [61, 83, 90, 97]. The second group of work, namely the Physics-Informed Neural Networks (PINNs)[15, 40, 42, 52, 77, 87, 96, 106, 127], treat neural networks as a parametrization of the underlying solution function. PINNs incorporate the knowledge of the governing equations into the boundary of loss function, which includes the residual of the PDE, the consistency with given boundary condition and initial condition. Unlike the previous group of works, PINNs do not necessarily need input-target data and can be trained solely based on equation loss. The third group of works, often referred to as the neural operator, focuses on learning a mapping between the function spaces[5, 8, 9, 16, 36, 43, 50, 54, 60, 68, 70, 71, 76, 78, 86]. Neural operator has the generalization capability within a family of PDE and can potentially be adapted to different discretization without retraining. DeepONet [76] proposes a practical realization of the universal operator approximation theorem [21]. Concurrent work graph neural operator [69] proposes a learnable kernel integral to approximate the solution operator of parametric PDEs and the follow-up work Fourier Neural Operator (FNO) [68] achieves excellent accuracy and efficiency on certain types of problems. Broadly speaking, the

OR

Candidate 2

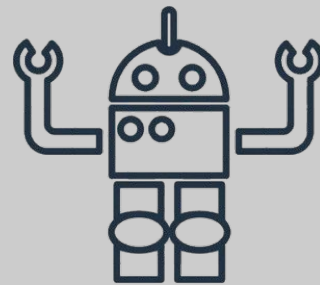
Transformer has shown state-of-the-art performance on various applications and has recently emerged as a promising tool for surrogate modeling of partial differential equations (PDEs). Despite the introduction of linear-complexity attention, applying Transformer to problems with a large number of grid points can be numerically unstable and computationally expensive. In this work, we propose Factorized Transformer (FactFormer), which is based on an axial factorized kernel integral. Concretely, we introduce a learnable projection operator that decomposes the input function into multiple sub-functions with one-dimensional domain. These sub-functions are then evaluated and used to compute the instance-based kernel with an axial factorized scheme. We showcase that the proposed model is able to simulate 2D Kolmogorov flow on a 256×256 grid and 3D smoke buoyancy on a $64 \times 64 \times 64$ grid with good accuracy and efficiency. The proposed factorized scheme can serve as a computationally efficient low-rank surrogate for the full attention scheme when dealing with multi-dimensional problems.

Human



Candidate 1 is fine-grain match

VLM



Candidate 1: 0
Candidate 2: 0.85
Choice: Candidate 2